

SOLIDWORKS API (.NET VBA C#) — Reference (119/337): IFeatureManager (part 6 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 6 of 9) (Type)

SOLIDWORKS API Help

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InsertFreeform Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFreeform Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertFreeform2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertFreeform( _  
    ByVal Continuity0 As System.Short, _  
    ByVal Continuity1 As System.Short, _  
    ByVal Continuity2 As System.Short, _  
    ByVal Continuity3 As System.Short, _  
    ByVal Symmetric0 As System.Boolean, _  
    ByVal Symmetric1 As System.Boolean _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Continuity0 As System.Short  
Dim Continuity1 As System.Short  
Dim Continuity2 As System.Short  
Dim Continuity3 As System.Short  
Dim Symmetric0 As System.Boolean  
Dim Symmetric1 As System.Boolean  
Dim value As Feature  
  
value = instance.InsertFreeform(Continuity0, Continuity1, Continuity2, Continuity3, Symmetric0,  
Symmetric1)
```

C#

```
Feature InsertFreeform(  
    System.short Continuity0,  
    System.short Continuity1,  
    System.short Continuity2,  
    System.short Continuity3,  
    System.bool Symmetric0,  
    System.bool Symmetric1  
)
```

C++/CLI

```
Feature^ InsertFreeform(  
& System.short Continuity0,  
& System.short Continuity1,  
& System.short Continuity2,  
& System.short Continuity3,  
& System.bool Symmetric0,  
& System.bool Symmetric1  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Continuity0

Continuity of edge 1:

- -1 = Movable
- 0 = Contact
- 2 = Tangent
- 3 = Curvature

Continuity1

Continuity of edge 2:

- -1 = Movable

- 0 = Contact
- 2 = Tangent
- 3 = Curvature

Continuity2

Continuity of edge 3:

- -1 = Movable
- 0 = Contact
- 2 = Tangent
- 3 = Curvature

Continuity3

Continuity of edge 4:

- -1 = Movable
- 0 = Contact
- 2 = Tangent
- 3 = Curvature

Symmetric0

True if symmetry is on in Direction 0, false if not

Symmetric1

True if symmetry is on in Direction 1, false if not

Return Value

Pointer to the newly created freeform [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFreeform](#).

Remarks

The SOLIDWORKS API Freeform-related methods are intended to journal the actions performed by an interactive user while creating the feature. Because user interaction is required to create a Freeform feature, fully automating the creation of it is not possible using the SOLIDWORKS API.

The typical steps performed by an interactive user to create a Freeform feature are:

1. Select the face to form.

NOTE: The face must have four sides.

2. Add curves on the selected face. Corresponds to [IFeatureManager::SetFreeformCurveData](#).
3. Add points on the curves. Corresponds to [IFeatureManager::SetFreeformPointData](#).

Interactively pull or push the points to change the shape of the selected face.

4. Insert the Freeform feature. Corresponds to this method, [IFeatureManager::InsertFreeform](#).

Record a macro while interactively creating a Freeform feature, then examine the macro to see the order in which the Freeform-related methods are recorded.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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InsertFreeform2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertFreeform2 Method (IFeatureManager)

Inserts a Freeform feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertFreeform2( _
    ByVal Nsided As System.Boolean, _
    ByVal Symmetric0 As System.Boolean, _
    ByVal Symmetric1 As System.Boolean, _
```

```
ByVal Angle As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Nsided As System.Boolean  
Dim Symmetric0 As System.Boolean  
Dim Symmetric1 As System.Boolean  
Dim Angle As System.Double  
Dim value As Feature  
  
value = instance.InsertFreeform2(Nsided, Symmetric0, Symmetric1, Angle)
```

C#

```
Feature InsertFreeform2(  
    System.bool Nsided,  
    System.bool Symmetric0,  
    System.bool Symmetric1,  
    System.double Angle  
)
```

C++/CLI

```
Feature^ InsertFreeform2(  
& System.bool Nsided,  
& System.bool Symmetric0,  
& System.bool Symmetric1,  
& System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Nsided

True to use the new Freeform algorithm, false to use the old one

NOTE: Must be true for faces with three sides or more than four sides, or to enable rotation of UV lines

Symmetric0

True if symmetry is on in Direction 0, false if not

Symmetric1

True if symmetry is on in Direction 1, false if not

Angle

Value by which to rotate the U,V lines (in radians)

Return Value

[Freeform feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertFreeform2](#).

Remarks

The SOLIDWORKS API Freeform-related methods are intended to journal the actions performed by an interactive user while creating the feature. Because user interaction is required to create a Freeform feature, fully automating the creation of it is not possible using the SOLIDWORKS API.

The typical steps performed by an interactive user to create a Freeform feature are:

1. Select the face to form.
2. Add curves on the selected face. Corresponds to [IFeatureManager::SetFreeformCurveData](#).
3. Add points on the curves. Corresponds to [IFeatureManager::SetFreeformPointData](#).
4. Specify boundary continuity. Corresponds to [IFeatureManager::SetFreeformBoundaryContinuity](#).

Interactively pull or push the points to change the shape of the selected face.

5. Insert the Freeform feature. Corresponds to this method, [IFeatureManager::InsertFreeform2](#).

Record a macro while interactively creating a Freeform feature, then examine the macro to see the order in which the Freeform-related methods are recorded.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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InsertGlobalBoundingBox Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGlobalBoundingBox Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateDefinition](#) and [IBoundingBoxFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGlobalBoundingBox( _  
    ByVal BBoxType As System.Integer, _  
    ByVal IncludeHiddenBodies As System.Boolean, _  
    ByVal IncludeSurfaceBodies As System.Boolean, _  
    ByRef Status As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BBoxType As System.Integer  
Dim IncludeHiddenBodies As System.Boolean  
Dim IncludeSurfaceBodies As System.Boolean  
Dim Status As System.Integer  
Dim value As System.Object
```

```
value = instance.InsertGlobalBoundingBox(BBoxType, IncludeHiddenBodies, IncludeSurfaceBodies, Status)
```

C#

```
System.object InsertGlobalBoundingBox(  
    System.int BBoxType,  
    System.bool IncludeHiddenBodies,  
    System.bool IncludeSurfaceBodies,  
    out System.int Status  
)
```

C++/CLI

```
System.Object^ InsertGlobalBoundingBox(  
& System.int BBoxType,  
& System.bool IncludeHiddenBodies,  
& System.bool IncludeSurfaceBodies,  
& [Out] System.int Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BBoxType

Bounding Box fit type as defined in [swGlobalBoundingBoxFitOptions_e](#) (see **Remarks**)

IncludeHiddenBodies

True to include hidden bodies, false to not

IncludeSurfaceBodies

True to include surfaces, false to not

Status

Status as defined by [swGlobalBoundingBoxResult_e](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertGlobalBoundingBox](#).

Example

' VBA

```

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As Boolean
Dim longstatus As Long
Option Explicit

Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    ' Display the Bounding Box sketch
    boolstatus = Part.SetUserPreferenceToggle(swViewDispGlobalBBox, True)

    Dim BoundingBox As SldWorks.Feature
    Set BoundingBox = Part.FeatureManager.InsertGlobalBoundingBox(swBoundingBoxType_BestFit, True,
False, longstatus)

    Part.ClearSelection2 True

    ' Hide the Bounding Box sketch
    boolstatus = Part.SetUserPreferenceToggle(swViewDispGlobalBBox, False)

End Sub

```

Remarks

If BBoxType is set to swGlobalBoundingBoxFitOptions_e.swBoundingBoxType_CustomPlane, then select a face or plane using [IModelDocExtension::SelectByRay](#) with TypeWanted set to [swSelectType_e.swSelFACES](#) before calling this method.

To display or hide the Bounding Box sketch, call [IModelDoc2::SetUserPreferenceToggle](#) to set [swUserPreferenceToggle_e.swViewDispGlobalBBox](#) to true or false, respectively.

After calling this method, use [IModelDoc2::ClearSelection2](#) to clear the selection made when the Bounding Box is created.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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InsertGlobalBoundingBox2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGlobalBoundingBox2 Method (IFeatureManager)

Inserts the specified global bounding box using a custom coordinate system. See [IFeatureManager::CreateDefinition](#) and [IBoundingBoxFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertGlobalBoundingBox2( _
    ByVal BBoxType As System.Integer, _
    ByVal lwtadirectionType As System.Integer, _
    ByVal IncludeHiddenBodies As System.Boolean, _
    ByVal IncludeSurfaceBodies As System.Boolean, _
    ByRef Status As System.Integer _
) As System.Object

```

Visual Basic (Usage)

```

Dim instance As IFeatureManager
Dim BBoxType As System.Integer
Dim lwtadirectionType As System.Integer
Dim IncludeHiddenBodies As System.Boolean
Dim IncludeSurfaceBodies As System.Boolean
Dim Status As System.Integer

```



Copy Code

```
Dim value As System.Object
```

```
value = instance.InsertGlobalBoundingBox2(BBoxType, lwtadirectionType, IncludeHiddenBodies,  
IncludeSurfaceBodies, Status)
```

C#

```
System.Object InsertGlobalBoundingBox2(  
    System.int BBoxType,  
    System.int lwtadirectionType,  
    System.bool IncludeHiddenBodies,  
    System.bool IncludeSurfaceBodies,  
    out System.int Status  
)
```

C++/CLI

```
System.Object^ InsertGlobalBoundingBox2(  
& System.int BBoxType,  
& System.int lwtadirectionType,  
& System.bool IncludeHiddenBodies,  
& System.bool IncludeSurfaceBodies,  
& [Out] System.int Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BBoxType
Bounding Box fit type as defined in [swGlobalBoundingBoxFitOptions_e](#) (see **Remarks**)
lwtadirectionType
Direction type as defined by [swGlobalBBoxLWTADirectionType_e](#)
IncludeHiddenBodies
True to include hidden bodies, false to not
IncludeSurfaceBodies
True to include surfaces, false to not
Status
Status as defined by [swGlobalBoundingBoxResult_e](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertGlobalBoundingBox2](#).

Remarks

If BBoxType is set to swGlobalBoundingBoxFitOptions_e.swBoundingBoxType_CustomPlane, then select a face or plane using [IModelDocExtension.SelectByRay](#) with TypeWanted set to [swSelectType_e.swSelFACES](#) before calling this method.

To display or hide the Bounding Box sketch, call [IModelDoc2.SetUserPreferenceToggle](#) to set [swUserPreferenceToggle_e.swViewDispGlobalBBox](#) to true or false, respectively.

After calling this method, use [IModelDoc2.ClearSelection2](#) to clear the selection made when the Bounding Box is created.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertGlobalBoundingBox Method \(\)](#)
[IBoundingBoxFeatureData Interface](#)

Availability

SOLIDWORKS 2026 SP01, Revision Number 34.1

SOLIDWORKS API Help

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InsertGridFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGridFeature Method (IFeatureManager)

Inserts a Grid System feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGridFeature( _  
    ByVal HeightDoubles As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim HeightDoubles As System.Object  
Dim value As Feature  
  
value = instance.InsertGridFeature(HeightDoubles)
```

C#

```
Feature InsertGridFeature(  
    System.object HeightDoubles  
)
```

C++/CLI

```
Feature^ InsertGridFeature(  
    & System.Object^ HeightDoubles  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

HeightDoubles

Array of doubles specifying distances between Grid System levels

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertGridFeature](#).

Example

[Insert Grid System Feature \(C#\)](#)
[Insert Grid System Feature \(VB.NET\)](#)
[Insert Grid System Feature \(VBA\)](#)

Remarks

The number of elements in HeightDoubles determines the number of levels in this Grid System feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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InsertGroundPlane Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGroundPlane Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateDefinition](#) and [IGroundPlaneFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGroundPlane( _  
    ByVal ReverseDirection As System.Boolean _
```

) As Feature

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim ReverseDirection As System.Boolean  
Dim value As Feature  
  
value = instance.InsertGroundPlane(ReverseDirection)
```

C#

```
Feature InsertGroundPlane(  
    System.bool ReverseDirection  
)
```

C++/CLI

```
Feature^ InsertGroundPlane(  
    & System.bool ReverseDirection  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReverseDirection

True to reverse the alignment of ground faces relative to the ground plane, false to not

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertGroundPlane](#).

Example

[Insert and Activate Ground Plane \(VBA\)](#)

Remarks

Before calling this method, select the entity to use as the ground plane using [IModelDocExtension::SelectByRay](#) or [IModelDocExtension::SelectByID2](#).

Call this method to insert a ground plane at each level of a plant assembly. Ground planes are used as reference geometry when inserting published assets at each level.

Call [IAssemblyDoc::ActivateGroundPlane](#) to activate the ground plane on the level where you want to insert a published asset; the asset's ground face snaps to the assembly's activated ground plane. Components with magnetic mates snap only to components that are also on the active ground plane.

Only one ground plane can be active at a given time in each assembly configuration.

See **Large Assemblies > Facility Layout** in the SOLIDWORKS Design user-interface help.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IAssemblyDoc::GetActiveGroundPlane Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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InsertGussetFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGussetFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertGussetFeature3](#).

.NET Syntax

```

Function InsertGussetFeature( _
    ByVal Depth As System.Double, _
    ByVal DirType As System.Short, _
    ByVal LocType As System.Short, _
    ByVal BIsProfile As System.Boolean, _
    ByVal ProfileD1 As System.Double, _
    ByVal ProfileD2 As System.Double, _
    ByVal ProfileD3 As System.Double, _
    ByVal ProfileAngle As System.Double, _
    ByVal ProfileD4 As System.Double, _
    ByVal BOffset As System.Boolean, _
    ByVal DProfileOffset As System.Double, _
    ByVal CrvIndex As System.Integer, _
    ByVal BReverseDir As System.Boolean, _
    ByVal BReverseFace As System.Boolean, _
    ByVal BUseLenDim As System.Boolean _
) As Feature

```

```

Dim instance As IFeatureManager
Dim Depth As System.Double
Dim DirType As System.Short
Dim LocType As System.Short
Dim BIsProfile As System.Boolean
Dim ProfileD1 As System.Double
Dim ProfileD2 As System.Double
Dim ProfileD3 As System.Double
Dim ProfileAngle As System.Double
Dim ProfileD4 As System.Double
Dim BOffset As System.Boolean
Dim DProfileOffset As System.Double
Dim CrvIndex As System.Integer
Dim BReverseDir As System.Boolean
Dim BReverseFace As System.Boolean
Dim BUseLenDim As System.Boolean
Dim value As Feature

```

```

value = instance.InsertGussetFeature(Depth, DirType, LocType, BIsProfile, ProfileD1, ProfileD2,
ProfileD3, ProfileAngle, ProfileD4, BOffset, DProfileOffset, CrvIndex, BReverseDir, BReverseFace,
BUseLenDim)

```

```

Feature InsertGussetFeature(
    System.double Depth,
    System.short DirType,
    System.short LocType,
    System.bool BIsProfile,
    System.double ProfileD1,
    System.double ProfileD2,
    System.double ProfileD3,
    System.double ProfileAngle,
    System.double ProfileD4,
    System.bool BOffset,
    System.double DProfileOffset,
    System.int CrvIndex,
    System.bool BReverseDir,
    System.bool BReverseFace,
    System.bool BUseLenDim
)

```

```

Feature^ InsertGussetFeature(
    & System.double Depth,
    & System.short DirType,
    & System.short LocType,
    & System.bool BIsProfile,
    & System.double ProfileD1,
    & System.double ProfileD2,

```

```

& System.double ProfileD3,
& System.double ProfileAngle,
& System.double ProfileD4,
& System.bool BOffset,
& System.double DProfileOffset,
& System.int CrvIndex,
& System.bool BReverseDir,
& System.bool BReverseFace,
& System.bool BUseLenDim
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

Depth of the gusset

DirType

Direction in which to extrude the gusset:

- 0 = left side
- 1 = center
- 2 = right side

LocType

Location of the reference plane for the sketch of the gusset:

- 0 = start point
- 1 = midpoint
- 2 = end point

BlsProfile

True to use a polygon profile, false to use a triangle

ProfileD1

Length for [Direction 1](#)

ProfileD2

Length for [Direction 2](#)

ProfileD3

Length for [Direction 3](#)

ProfileAngle

Value for [profile angle](#)

ProfileD4

Length for [Direction 4](#)

BOffset

True to offset the reference plane for the sketch, false to not

DProfileOffset

Value by which to offset the reference plane for the sketch

CrvIndex

Index of the edge to use if multiple intersecting edges exist

BReverseDir

If BOffset set to true, then true to reverse direction, false to not

BReverseFace

Reverse ProfileD1 and ProfileD2 if triangle profile

- or -

Reverse ProfileD1 and ProfileD2 and reverse ProfileD3 and ProfileD4 if polygon profile

BUseLenDim

True to use ProfileD4, false to use ProfileAngle

Return Value

Pointer to the newly created [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertGussetFeature](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IGussetFeatureData Interface](#)

[IFeatureManager::InsertGussetFeature2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertGussetFeature2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGussetFeature2 Method (IFeatureManager)

Inserts a gusset feature for the specified faces.

Visual Basic (Declaration)

```
Function InsertGussetFeature2( _
    ByVal Depth As System.Double, _
    ByVal DirType As System.Short, _
    ByVal LocType As System.Short, _
    ByVal BIsProfile As System.Boolean, _
    ByVal ProfileD1 As System.Double, _
    ByVal ProfileD2 As System.Double, _
    ByVal ProfileD3 As System.Double, _
    ByVal ProfileAngle As System.Double, _
    ByVal ProfileD4 As System.Double, _
    ByVal BOffset As System.Boolean, _
    ByVal DProfileOffset As System.Double, _
    ByVal CrvIndex As System.Integer, _
    ByVal BReverseDir As System.Boolean, _
    ByVal BReverseFace As System.Boolean, _
    ByVal BUseLenDim As System.Boolean, _
    ByVal Faces As System.Object _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Depth As System.Double
Dim DirType As System.Short
Dim LocType As System.Short
Dim BIsProfile As System.Boolean
Dim ProfileD1 As System.Double
Dim ProfileD2 As System.Double
Dim ProfileD3 As System.Double
Dim ProfileAngle As System.Double
Dim ProfileD4 As System.Double
Dim BOffset As System.Boolean
Dim DProfileOffset As System.Double
Dim CrvIndex As System.Integer
Dim BReverseDir As System.Boolean
Dim BReverseFace As System.Boolean
Dim BUseLenDim As System.Boolean
Dim Faces As System.Object
Dim value As Feature
```

```
value = instance.InsertGussetFeature2(Depth, DirType, LocType, BIsProfile, ProfileD1, ProfileD2,
ProfileD3, ProfileAngle, ProfileD4, BOffset, DProfileOffset, CrvIndex, BReverseDir, BReverseFace,
BUseLenDim, Faces)
```

C#

```
Feature InsertGussetFeature2(
    System.double Depth,
    System.short DirType,
    System.short LocType,
    System.bool BIsProfile,
    System.double ProfileD1,
    System.double ProfileD2,
    System.double ProfileD3,
    System.double ProfileAngle,
    System.double ProfileD4,
    System.bool BOffset,
    System.double DProfileOffset,
    System.int CrvIndex,
    System.bool BReverseDir,
    System.bool BReverseFace,
    System.bool BUseLenDim,
    System.object Faces
)
```

C++/CLI

```

Feature^ InsertGussetFeature2(
& System.double Depth,
& System.short DirType,
& System.short LocType,
& System.bool BIsProfile,
& System.double ProfileD1,
& System.double ProfileD2,
& System.double ProfileD3,
& System.double ProfileAngle,
& System.double ProfileD4,
& System.bool BOffset,
& System.double DProfileOffset,
& System.int CrvIndex,
& System.bool BReverseDir,
& System.bool BReverseFace,
& System.bool BUseLenDim,
& System.Object^ Faces
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

Depth of the gusset

DirType

Direction in which to extrude the gusset:

- 0 = left side
- 1 = center
- 2 = right side

LocType

Location of the reference plane for the sketch of the gusset:

- 0 = start point
- 1 = midpoint
- 2 = end point

BIsProfile

True to use a polygon profile, false to use a triangle

ProfileD1

Length for [Direction 1](#)

ProfileD2

Length for [Direction 2](#)

ProfileD3

Length for [Direction 3](#)

ProfileAngle

Value for [profile angle](#)

ProfileD4

Length for [Direction 4](#)

BOffset

True to offset the reference plane for the sketch, false to not

DProfileOffset

Value by which to offset the reference plane for the sketch

CrvIndex

Index of the edge to use if multiple intersecting edges exist

BReverseDir

If BOffset set to true, then true to reverse direction, false to not

BReverseFace

Reverse ProfileD1 and ProfileD2 if triangle profile

- or -

Reverse ProfileD1 and ProfileD2 and reverse ProfileD3 and ProfileD4 if polygon profile

BUseLenDim

True to use ProfileD4, false to use ProfileAngle

Faces

Array of [faces](#) that form the bases of this gusset feature

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertGussetFeature2](#).

Example

[Insert Weldment Features \(VBA\)](#)
[Insert Weldment Features \(VB.NET\)](#)
[Insert Weldment Features \(C#\)](#)

Remarks

Instead of using this method, you can:

1. Select two faces in the graphics area that are the supporting legs of this gusset.
2. Call [IFeatureManager.InsertGussetFeature3](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IGussetFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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InsertGussetFeature3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertGussetFeature3 Method (IFeatureManager)

Inserts a gusset feature for pre-selected faces of a weldment.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGussetFeature3( _  
    ByVal Depth As System.Double, _  
    ByVal DirType As System.Short, _  
    ByVal LocType As System.Short, _  
    ByVal BIsProfile As System.Boolean, _  
    ByVal ProfileD1 As System.Double, _  
    ByVal ProfileD2 As System.Double, _  
    ByVal ProfileD3 As System.Double, _  
    ByVal ProfileAngle As System.Double, _  
    ByVal ProfileD4 As System.Double, _  
    ByVal BOffset As System.Boolean, _  
    ByVal DProfileOffset As System.Double, _  
    ByVal CrvIndex As System.Integer, _  
    ByVal BReverseDir As System.Boolean, _  
    ByVal BReverseFace As System.Boolean, _  
    ByVal BUseLenDim As System.Boolean, _  
    ByVal ProfileD5 As System.Double, _  
    ByVal ProfileD6 As System.Double, _  
    ByVal ProfileChamferAngle As System.Double, _  
    ByVal BUseLenDimForChamfer As System.Boolean, _  
    ByVal BNeedChamferInGusset As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Depth As System.Double  
Dim DirType As System.Short  
Dim LocType As System.Short  
Dim BIsProfile As System.Boolean  
Dim ProfileD1 As System.Double  
Dim ProfileD2 As System.Double  
Dim ProfileD3 As System.Double  
Dim ProfileAngle As System.Double  
Dim ProfileD4 As System.Double  
Dim BOffset As System.Boolean  
Dim DProfileOffset As System.Double  
Dim CrvIndex As System.Integer  
Dim BReverseDir As System.Boolean  
Dim BReverseFace As System.Boolean  
Dim BUseLenDim As System.Boolean  
Dim ProfileD5 As System.Double  
Dim ProfileD6 As System.Double  
Dim ProfileChamferAngle As System.Double  
Dim BUseLenDimForChamfer As System.Boolean  
Dim BNeedChamferInGusset As System.Boolean
```



Dim value As Feature

```
value = instance.InsertGussetFeature3(Depth, DirType, LocType, BIsProfile, ProfileD1, ProfileD2, ProfileD3, ProfileAngle, ProfileD4, BOffset, DProfileOffset, CrvIndex, BReverseDir, BReverseFace, BUseLenDim, ProfileD5, ProfileD6, ProfileChamferAngle, BUseLenDimForChamfer, BNeedChamferInGusset)
```

C#

```
Feature InsertGussetFeature3(  
    System.double Depth,  
    System.short DirType,  
    System.short LocType,  
    System.bool BIsProfile,  
    System.double ProfileD1,  
    System.double ProfileD2,  
    System.double ProfileD3,  
    System.double ProfileAngle,  
    System.double ProfileD4,  
    System.bool BOffset,  
    System.double DProfileOffset,  
    System.int CrvIndex,  
    System.bool BReverseDir,  
    System.bool BReverseFace,  
    System.bool BUseLenDim,  
    System.double ProfileD5,  
    System.double ProfileD6,  
    System.double ProfileChamferAngle,  
    System.bool BUseLenDimForChamfer,  
    System.bool BNeedChamferInGusset  
)
```

C++/CLI

```
Feature^ InsertGussetFeature3(  
& System.double Depth,  
& System.short DirType,  
& System.short LocType,  
& System.bool BIsProfile,  
& System.double ProfileD1,  
& System.double ProfileD2,  
& System.double ProfileD3,  
& System.double ProfileAngle,  
& System.double ProfileD4,  
& System.bool BOffset,  
& System.double DProfileOffset,  
& System.int CrvIndex,  
& System.bool BReverseDir,  
& System.bool BReverseFace,  
& System.bool BUseLenDim,  
& System.double ProfileD5,  
& System.double ProfileD6,  
& System.double ProfileChamferAngle,  
& System.bool BUseLenDimForChamfer,  
& System.bool BNeedChamferInGusset  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

Thickness of the gusset

DirType

Thickness direction of the gusset as defined in [swGussetThicknessType_e](#)

LocType

Location of the reference plane for the sketch of the gusset as defined in [swGussetProfileLocationType_e](#)

BIsProfile

True to use a polygon profile, false to use a triangle profile

ProfileD1

Profile Distance1 (see **Remarks**)

ProfileD2

Profile Distance2 (see **Remarks**)

ProfileD3

Profile Distance3 (see **Remarks**)

ProfileAngle

Profile Angle (see **Remarks**); valid only if BUseLenDim is false

ProfileD4

Profile Distance4 (see **Remarks**); valid only if BUseLenDim is true

BOffset

True to offset the reference plane for the sketch, false to not

DProfileOffset
Value by which to offset the reference plane for the sketch
CrvIndex
Index of the edge to use if multiple intersecting edges exist
BReverseDir
If BOffset set to true, then true to reverse direction, false to not
BReverseFace
Reverse ProfileD1 and ProfileD2 if triangle profile

- Or -

Reverse ProfileD1 and ProfileD2 and reverse ProfileD3 and ProfileD4 if polygon profile

BUseLenDim
True to use ProfileD4, false to use ProfileAngle
ProfileD5
Chamfer Distance5 (see **Remarks**); valid only if BNeedChamferInGusset is true
ProfileD6
Chamfer Distance6 (see **Remarks**); valid only if BNeedChamferInGusset and BUseLenDimForChamber are both true
ProfileChamferAngle
Chamfer Angle (see **Remarks**); valid only if BUseLenDimForChamber is false and BNeedChamferInGusset is true
BUseLenDimForChamber
True to use ProfileD6, false to use ProfileChamferAngle; valid only if BNeedChamferInGusset is true
BNeedChamferInGusset
True to create a chamfer to allow room for a weld bead under the gusset, false to not

Return Value

[IFeature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertGussetFeature3](#).

 Example

[Insert Gusset Feature \(VBA\)](#)
[Insert Gusset Feature \(VB.NET\)](#)
[Insert Gusset Feature \(C#\)](#)

 Remarks

See the SOLIDWORKS Help for more information about profile and chamfer distances and angles in weldment gussets.

Before calling this method, you must call [IModelDocExtension::SelectByID2](#) to select two faces that are the supporting legs of this gusset with a Mark c 1.

Instead of using this method, you can pass an array of faces to [IFeatureManager::InsertGussetFeature2](#).

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IGussetFeatureData Interface](#)

 Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertIndent Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertIndent Method (IFeatureManager)

Inserts an indent feature using a selected target body and tool body regions.

 .NET Syntax

Visual Basic (Declaration)

```
Function InsertIndent( _  
    ByVal Thickness As System.Double, _  
    ByVal Clearance As System.Double, _  
    ByVal Exclude As System.Boolean, _  
    ByVal ClrDir As System.Boolean, _  
    ByVal Cut As System.Boolean, _  
    ByVal CutDir As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Thickness As System.Double
Dim Clearance As System.Double
Dim Exclude As System.Boolean
Dim ClrDir As System.Boolean
Dim Cut As System.Boolean
Dim CutDir As System.Boolean
Dim value As Feature

```

```
value = instance.InsertIndent(Thickness, Clearance, Exclude, ClrDir, Cut, CutDir)
```

C#

```

Feature InsertIndent(
    System.double Thickness,
    System.double Clearance,
    System.bool Exclude,
    System.bool ClrDir,
    System.bool Cut,
    System.bool CutDir
)

```

C++/CLI

```

Feature^ InsertIndent(
    & System.double Thickness,
    & System.double Clearance,
    & System.bool Exclude,
    & System.bool ClrDir,
    & System.bool Cut,
    & System.bool CutDir
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Thickness of the indent feature

Clearance

Distance between the tool body and target body (see **Remarks**)

Exclude

True to exclude the selections, false to include the selections

ClrDir

True to leave the direction of clearance as is, false to reverse the direction of the clearance

Cut

True to cut the target body, false to not

CutDir

True to reverse the direction of the cut if the tool body is a surface, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertIndent](#).

Example

[Insert Indent Feature \(C#\)](#)

[Insert Indent Feature \(VB.NET\)](#)

[Insert Indent Feature \(VBA\)](#)

Remarks

Prior to calling this method, you must have selected the target body and tool body regions, using these selection marks:

- Target body = 1
- Tool body region = 4

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IndentFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertLiveSectionPlane Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertLiveSectionPlane Method (IFeatureManager)

Inserts a Live Section Plane using the selected plane or planar face.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertLiveSectionPlane( _  
    ByVal Type As System.Short _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Short  
Dim value As Feature  
  
value = instance.InsertLiveSectionPlane(Type)
```

C#

```
Feature InsertLiveSectionPlane(  
    System.short Type  
)
```

C++/CLI

```
Feature^ InsertLiveSectionPlane(  
    & System.short Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type
Not used. Specify any integer value.

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertLiveSectionPlane](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.MoveRotateLiveSectionPlane Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMacroFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMacroFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMacroFeature3](#).

Visual Basic (Declaration)

```
Function InsertMacroFeature( _
    ByVal BaseName As System.String, _
    ByVal ProgId As System.String, _
    ByVal MacroMethods As System.Object, _
    ByVal ParamNames As System.Object, _
    ByVal ParamTypes As System.Object, _
    ByVal ParamValues As System.Object, _
    ByVal EditBody As Body2, _
    ByVal Options As System.Integer _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim BaseName As System.String
Dim ProgId As System.String
Dim MacroMethods As System.Object
Dim ParamNames As System.Object
Dim ParamTypes As System.Object
Dim ParamValues As System.Object
Dim EditBody As Body2
Dim Options As System.Integer
Dim value As Feature

value = instance.InsertMacroFeature(BaseName, ProgId, MacroMethods, ParamNames, ParamTypes,
ParamValues, EditBody, Options)
```

C#

```
Feature InsertMacroFeature(
    System.string BaseName,
    System.string ProgId,
    System.object MacroMethods,
    System.object ParamNames,
    System.object ParamTypes,
    System.object ParamValues,
    Body2 EditBody,
    System.int Options
)
```

C++/CLI

```
Feature^ InsertMacroFeature(
    & System.String^ BaseName,
    & System.String^ ProgId,
    & System.Object^ MacroMethods,
    & System.Object^ ParamNames,
    & System.Object^ ParamTypes,
    & System.Object^ ParamValues,
    & Body2^ EditBody,
    & System.int Options
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseName

ProgId

MacroMethods

ParamNames

ParamTypes

ParamValues

EditBody

Options

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMacroFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMacroFeature2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMacroFeature2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMacroFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMacroFeature2( _  
    ByVal BaseName As System.String, _  
    ByVal ProgId As System.String, _  
    ByVal MacroMethods As System.Object, _  
    ByVal ParamNames As System.Object, _  
    ByVal ParamTypes As System.Object, _  
    ByVal ParamValues As System.Object, _  
    ByVal DimTypes As System.Object, _  
    ByVal DimValues As System.Object, _  
    ByVal EditBody As Body2, _  
    ByVal IconFiles As System.Object, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BaseName As System.String  
Dim ProgId As System.String  
Dim MacroMethods As System.Object  
Dim ParamNames As System.Object  
Dim ParamTypes As System.Object  
Dim ParamValues As System.Object  
Dim DimTypes As System.Object  
Dim DimValues As System.Object  
Dim EditBody As Body2  
Dim IconFiles As System.Object  
Dim Options As System.Integer  
Dim value As Feature
```

```
value = instance.InsertMacroFeature2(BaseName, ProgId, MacroMethods, ParamNames, ParamTypes,  
ParamValues, DimTypes, DimValues, EditBody, IconFiles, Options)
```

C#

```
Feature InsertMacroFeature2(  
    System.string BaseName,  
    System.string ProgId,  
    System.object MacroMethods,  
    System.object ParamNames,  
    System.object ParamTypes,  
    System.object ParamValues,  
    System.object DimTypes,  
    System.object DimValues,  
    Body2 EditBody,  
    System.object IconFiles,  
    System.int Options  
)
```

C++/CLI

```
Feature^ InsertMacroFeature2(  
& System.String^ BaseName,  
& System.String^ ProgId,  
& System.Object^ MacroMethods,  
& System.Object^ ParamNames,  
& System.Object^ ParamTypes,  
& System.Object^ ParamValues,  
& System.Object^ DimTypes,  
& System.Object^ DimValues,  
& Body2^ EditBody,  
& System.Object^ IconFiles,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseName

ProgId

MacroMethods

ParamNames

ParamTypes

ParamValues

DimTypes

DimValues

EditBody

IconFiles

Options

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMacroFeature2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMacroFeature3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMacroFeature3 Method (IFeatureManager)

Inserts a macro feature in this model.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMacroFeature3( _  
    ByVal BaseName As System.String, _  
    ByVal ProgId As System.String, _  
    ByVal MacroMethods As System.Object, _  
    ByVal ParamNames As System.Object, _  
    ByVal ParamTypes As System.Object, _  
    ByVal ParamValues As System.Object, _  
    ByVal DimTypes As System.Object, _  
    ByVal DimValues As System.Object, _  
    ByVal EditBodies As System.Object, _  
    ByVal IconFiles As System.Object, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



Copy Code

```

Dim instance As IFeatureManager
Dim BaseName As System.String
Dim ProgId As System.String
Dim MacroMethods As System.Object
Dim ParamNames As System.Object
Dim ParamTypes As System.Object
Dim ParamValues As System.Object
Dim DimTypes As System.Object
Dim DimValues As System.Object
Dim EditBodies As System.Object
Dim IconFiles As System.Object
Dim Options As System.Integer
Dim value As Feature

```

```

value = instance.InsertMacroFeature3(BaseName, ProgId, MacroMethods, ParamNames, ParamTypes,
ParamValues, DimTypes, DimValues, EditBodies, IconFiles, Options)

```

C#

```

Feature InsertMacroFeature3(
    System.string BaseName,
    System.string ProgId,
    System.object MacroMethods,
    System.object ParamNames,
    System.object ParamTypes,
    System.object ParamValues,
    System.object DimTypes,
    System.object DimValues,
    System.object EditBodies,
    System.object IconFiles,
    System.int Options
)

```

C++/CLI

```

Feature^ InsertMacroFeature3(
    & System.String^ BaseName,
    & System.String^ ProgId,
    & System.Object^ MacroMethods,
    & System.Object^ ParamNames,
    & System.Object^ ParamTypes,
    & System.Object^ ParamValues,
    & System.Object^ DimTypes,
    & System.Object^ DimValues,
    & System.Object^ EditBodies,
    & System.Object^ IconFiles,
    & System.int Options
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseName

Name of the base feature (see **Remarks**)

ProgId

COM or .NET callback object; empty string if VBA (see **Remarks**)

MacroMethods

Array of strings of size 9 only for VBA; "" otherwise (see **Remarks**)

ParamNames

Array of strings of the names of the parameters

ParamTypes

Array of the types of parameters of size paramCount as defined by [swMacroFeatureParamType_e](#)

ParamValues

Array of strings of the values of parameters

DimTypes

Array of the types of dimensions as defined by [swDimensionType_e](#) (see **Remarks**)

DimValues

Array of values of the dimensions

EditBodies

Array of [IBody2](#) objects to modify in the macro feature

IconFiles

Array of strings of the files names for the icons (see **Remarks**)

Options

Placement of the macro feature in the FeatureManager design tree as defined by [swMacroFeatureOptions_e](#) (see **Remarks**)

Return Value

[IFeature](#)

See [FeatureManager::InsertMacroFeature3](#).

 Example

[Create Macro Feature Subfeature \(VBA\)](#)
[Create Multibody Macro Feature \(VBA\)](#)
[Create Multibody Macro Feature \(VB.NET\)](#)
[Create Multibody Macro Feature \(C#\)](#)

 Remarks

BaseName

- The argument BaseName is serialized within the feature and cannot be changed.
- You can find out the name of the base feature by using [IMacroFeatureData::GetBaseName](#).
- The BaseName argument is also used to generate the name of the feature when the feature is first created.

ProgId

If creating a macro feature using...	Then ProgId is...
COM callbacks	The name of the program ID for the component that implements the COM callback methods. The that is represented by ProgID must support the ISwComFeature interface. InsertMacroFeature3 ("Sample", "Sample.MyFeature", _ Nothing, paramNames, paramTypes, _ paramValues, dimTypes, dimValues, _ editBodies, iconFiles, _ swMacroFeatureByDefault) In the COM server, the Sample.Feature class is derived from ISwComFeature and implements Iedit, and security functions.
VBA	An empty string. InsertMacroFeature3("Sample", "",...)
.NET add-in callbacks	The name of the class that implements the .NET callbacks and ISwComFeature. InsertMacroFeature3("Sample", "project_name.class_module",...)

MacroMethods

The macroMethods argument is implemented for VBA only. The array of nine strings consists of the following values:

1. Filename - File executed during feature generation.
2. Module - Source module executed during feature generation.
3. Procedure - Source procedure executed during feature generation.
4. Filename - File executed after edit definition is selected.
5. Module - Source module executed after edit definition is selected.
6. Procedure - Source procedure executed after edit definition is selected.
7. Filename - File executed while querying security; optional, see the next paragraph.
8. Module - Source Module executed while querying security; optional, see the next paragraph.
9. Procedure - Source Procedure executed while querying security; optional, see the next paragraph.

Filename should be the full pathname to the macro file. If the procedure resides in the same macro file that calls IFeatureManager::InsertMacroFeature3, then a call to [ISldWorks::GetCurrentMacroPathName](#) provides all of the information necessary for the Filename.

[IModelDoc2::ListAuxiliaryExternalFileReferences](#) returns an array containing the names of the features that include external references and an array containing the names of the external files for Filename elements 1, 4, and 7 in the array.

The list of:

- features can contain duplicates, if the macro feature uses more than one procedure.
- file names can contain duplicates if:
 - the procedures are implemented in the same macro file.
 - more than one instance of the same macro feature is present.

If a security procedure is not used, then Filename, Module, and Procedure must all be empty strings.

Procedure names must have an **swm** prefix in the name. This prefix identifies the procedures to execute.

DimTypes

Only these dimension types are supported by this method:

- swAngularDimension
- swLinearDimension
- swRadialDimension

IconFiles

The array of the file names for the icons can contain either three or nine strings.

Number of strings in array	Types of images in this order	Image format and sizes
Three	1. Regular 2. Suppressed 3. Highlighted	• Windows bitmap (*.bmp) format • 16 pixels wide X 16 pixels high
Nine	1. Regular small 2. Suppressed small 3. Highlighted small 4. Regular medium 5. Suppressed medium 6. Highlighted medium 7. Regular large 8. Suppressed large 9. Highlighted large	• Windows bitmap (*.bmp) format • Recommended sizes are: • Small: 20 pixels wide X 20 pixels high • Medium: 32 pixels wide X 32 pixels high • Large: 40 pixels wide X 40 pixels high

NOTES:

- This size array is only valid for macro features created in parts, assemblies, and drawings in SOLIDWORKS 2017 and later.
- SOLIDWORKS displays the appropriate images based on the current DPI setting of the display device.

You can specify either the full path name or just the file name for the strings; for example, **c:\bitmaps\icon1.bmp** or **icon1.bmp**.

Options

swMacroFeatureOptions_e.swMacroFeatureEmbedMacroFile is not supported by programming languages for the Microsoft .NET Framework; for example, not supported by C#, Visual Basic .NET, or Managed C++.

See [Overview of Macro Features](#) for more information.

See Also

- [IFeatureManager Interface](#)
- [IFeatureManager Members](#)
- [IFeatureManager.InsertMacroFeature3 Method](#)
- [MacroFeatureData Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMateReference Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMateReference Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMateReference2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMateReference( _
    ByVal BstrMateReferenceName As System.String, _
    ByVal PrimaryReferenceEntity As Entity, _
    ByVal PrimaryReferenceType As System.Integer, _
    ByVal PrimaryReferenceAlignment As System.Integer, _
    ByVal SecondaryReferenceEntity As Entity, _
    ByVal SecondaryReferenceType As System.Integer, _
    ByVal SecondaryReferenceAlignment As System.Integer, _
    ByVal TertiaryReferenceEntity As Entity, _
    ByVal TertiaryReferenceType As System.Integer, _
    ByVal TertiaryReferenceAlignment As System.Integer _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim BstrMateReferenceName As System.String
Dim PrimaryReferenceEntity As Entity
Dim PrimaryReferenceType As System.Integer
Dim PrimaryReferenceAlignment As System.Integer
Dim SecondaryReferenceEntity As Entity
Dim SecondaryReferenceType As System.Integer
Dim SecondaryReferenceAlignment As System.Integer
Dim TertiaryReferenceEntity As Entity
Dim TertiaryReferenceType As System.Integer
Dim TertiaryReferenceAlignment As System.Integer
Dim value As Feature

value = instance.InsertMateReference(BstrMateReferenceName, PrimaryReferenceEntity,
PrimaryReferenceType, PrimaryReferenceAlignment, SecondaryReferenceEntity, SecondaryReferenceType,
SecondaryReferenceAlignment, TertiaryReferenceEntity, TertiaryReferenceType,
TertiaryReferenceAlignment)
```

C#

```
Feature InsertMateReference(  
    System.string BstrMateReferenceName,  
    Entity PrimaryReferenceEntity,  
    System.int PrimaryReferenceType,  
    System.int PrimaryReferenceAlignment,  
    Entity SecondaryReferenceEntity,  
    System.int SecondaryReferenceType,  
    System.int SecondaryReferenceAlignment,  
    Entity TertiaryReferenceEntity,  
    System.int TertiaryReferenceType,  
    System.int TertiaryReferenceAlignment  
)
```

C++/CLI

```
Feature^ InsertMateReference(  
& System.String^ BstrMateReferenceName,  
& Entity^ PrimaryReferenceEntity,  
& System.int PrimaryReferenceType,  
& System.int PrimaryReferenceAlignment,  
& Entity^ SecondaryReferenceEntity,  
& System.int SecondaryReferenceType,  
& System.int SecondaryReferenceAlignment,  
& Entity^ TertiaryReferenceEntity,  
& System.int TertiaryReferenceType,  
& System.int TertiaryReferenceAlignment  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BstrMateReferenceName
Name of mate reference
PrimaryReferenceEntity
Pointer to the primary mate, the [IEntity](#) object
PrimaryReferenceType
Primary mate's reference type as defined by [swMateReferenceType_e](#)
PrimaryReferenceAlignment
Primary mate's reference alignment type as defined [swMateReferenceAlignment_e](#)
SecondaryReferenceEntity
Pointer to the secondary mate, the [IEntity](#) object
SecondaryReferenceType
Secondary mate's reference type as defined by [swMateReferenceType_e](#)
SecondaryReferenceAlignment
Secondary mate's alignment type as defined [swMateReferenceAlignment_e](#)
TertiaryReferenceEntity
Pointer to the tertiary mate, the [IEntity](#) object
TertiaryReferenceType
Tertiary mate's reference type as defined by [swMateReferenceType_e](#)
TertiaryReferenceAlignment
Tertiary mate's reference alignment as defined by [swMateReferenceAlignment_e](#)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMateReference](#).

Remarks

Either select the mate references interactively or select them programmatically using [IModelDocExtension::SelectByID2](#) with marks of 1, 2, and 4. If the mate references are interactively selected, then specify NOTHING for PrimaryReferenceEntity, SecondaryReferenceEntity, and TertiaryReferenceEntity.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMateReference Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMateReference2 Method (IFeatureManager)

Inserts a mate reference for a part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMateReference2( _  
    ByVal BstrMateReferenceName As System.String, _  
    ByVal PrimaryReferenceEntity As Entity, _  
    ByVal PrimaryReferenceType As System.Integer, _  
    ByVal PrimaryReferenceAlignment As System.Integer, _  
    ByVal PrimaryReferenceAlignAxes As System.Boolean, _  
    ByVal SecondaryReferenceEntity As Entity, _  
    ByVal SecondaryReferenceType As System.Integer, _  
    ByVal SecondaryReferenceAlignment As System.Integer, _  
    ByVal SecondaryReferenceAlignAxes As System.Boolean, _  
    ByVal TertiaryReferenceEntity As Entity, _  
    ByVal TertiaryReferenceType As System.Integer, _  
    ByVal TertiaryReferenceAlignment As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BstrMateReferenceName As System.String  
Dim PrimaryReferenceEntity As Entity  
Dim PrimaryReferenceType As System.Integer  
Dim PrimaryReferenceAlignment As System.Integer  
Dim PrimaryReferenceAlignAxes As System.Boolean  
Dim SecondaryReferenceEntity As Entity  
Dim SecondaryReferenceType As System.Integer  
Dim SecondaryReferenceAlignment As System.Integer  
Dim SecondaryReferenceAlignAxes As System.Boolean  
Dim TertiaryReferenceEntity As Entity  
Dim TertiaryReferenceType As System.Integer  
Dim TertiaryReferenceAlignment As System.Integer  
Dim value As Feature  
  
value = instance.InsertMateReference2(BstrMateReferenceName, PrimaryReferenceEntity,  
PrimaryReferenceType, PrimaryReferenceAlignment, PrimaryReferenceAlignAxes, SecondaryReferenceEntity,  
SecondaryReferenceType, SecondaryReferenceAlignment, SecondaryReferenceAlignAxes,  
TertiaryReferenceEntity, TertiaryReferenceType, TertiaryReferenceAlignment)
```

C#

```
Feature InsertMateReference2(  
    System.string BstrMateReferenceName,  
    Entity PrimaryReferenceEntity,  
    System.int PrimaryReferenceType,  
    System.int PrimaryReferenceAlignment,  
    System.bool PrimaryReferenceAlignAxes,  
    Entity SecondaryReferenceEntity,  
    System.int SecondaryReferenceType,  
    System.int SecondaryReferenceAlignment,  
    System.bool SecondaryReferenceAlignAxes,  
    Entity TertiaryReferenceEntity,  
    System.int TertiaryReferenceType,  
    System.int TertiaryReferenceAlignment  
)
```

C++/CLI

```
Feature^ InsertMateReference2(  
& System.String^ BstrMateReferenceName,  
& Entity^ PrimaryReferenceEntity,  
& System.int PrimaryReferenceType,  
& System.int PrimaryReferenceAlignment,  
& System.bool PrimaryReferenceAlignAxes,  
& Entity^ SecondaryReferenceEntity,
```

```

& System.int SecondaryReferenceType,
& System.int SecondaryReferenceAlignment,
& System.bool SecondaryReferenceAlignAxes,
& Entity^ TertiaryReferenceEntity,
& System.int TertiaryReferenceType,
& System.int TertiaryReferenceAlignment
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BstrMateReferenceName
Name of mate reference

PrimaryReferenceEntity
Pointer to the primary mate, the [IEntity](#) object

PrimaryReferenceType
Primary mate's reference type as defined by [swMateReferenceType_e](#)

PrimaryReferenceAlignment
Primary mate's reference alignment type as defined [swMateReferenceAlignment_e](#)

PrimaryReferenceAlignAxes
True to align with the axes of the coordinate system or origin when forming a mate, false to not

SecondaryReferenceEntity
Pointer to the secondary mate, the [IEntity](#) object

SecondaryReferenceType
Secondary mate's reference type as defined by [swMateReferenceType_e](#)

SecondaryReferenceAlignment
Secondary mate's alignment type as defined [swMateReferenceAlignment_e](#)

SecondaryReferenceAlignAxes
True to align with the axes of the coordinate system or origin when forming a mate, false to not

TertiaryReferenceEntity
Pointer to the tertiary mate, the [IEntity](#) object

TertiaryReferenceType
Tertiary mate's reference type as defined by [swMateReferenceType_e](#)

TertiaryReferenceAlignment
Tertiary mate's reference alignment as defined by [swMateReferenceAlignment_e](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMateReference2](#).

Example

[Edit Mate Reference \(C#\)](#)
[Edit Mate Reference \(VB.NET\)](#)
[Edit Mate Reference \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMateReference Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMidSurface Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMidSurface Method (IFeatureManager)

Inserts a midsurface feature.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertMidSurface( _
    ByVal PBodyIn As Body2, _
    ByVal PPartDocIn As ModelDoc2, _
    ByVal Placement As System.Double, _
    ByVal KnitFlag As System.Boolean _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim PBodyIn As Body2
Dim PPartDocIn As ModelDoc2
Dim Placement As System.Double
Dim KnitFlag As System.Boolean
Dim value As System.Object

```

```

value = instance.InsertMidSurface(PBodyIn, PPartDocIn, Placement, KnitFlag)

```

C#

```

System.Object InsertMidSurface(
    Body2 PBodyIn,
    ModelDoc2 PPartDocIn,
    System.double Placement,
    System.bool KnitFlag
)

```

C++/CLI

```

System.Object^ InsertMidSurface(
    & Body2^ PBodyIn,
    & ModelDoc2^ PPartDocIn,
    & System.double Placement,
    & System.bool KnitFlag
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PBodyIn

[Body](#), in the context of an assembly; otherwise, null or Nothing (see **Remarks**)

PPartDocIn

[Model document](#), in the context of an assembly; otherwise, null or Nothing (see **Remarks**)

Placement

Value from -1 to 1 that indicates the desired location of the midsurface relative to the face pair; a value of 0.0 places the midsurface equally between the face pair

KnitFlag

True to sew all the reference surfaces (neutral sheets) into a single sheet (surface) body, false for the midsurface feature to contain individual reference surface geometry

Return Value

Newly created [midsurface feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMidSurface](#).

Example

[Insert MidSurface in Component \(C#\)](#)
[Insert MidSurface in Component \(VB.NET\)](#)
[Insert MidSurface in Component \(VBA\)](#)
[Get Areas of Midsurface Faces \(C#\)](#)
[Get Areas of Midsurface Faces \(VB.NET\)](#)
[Get Areas of Midsurface Faces \(VBA\)](#)

Remarks

The pPartDocIn parameter is the component's model document where you want to add the midsurface feature, and the pBody parameter is the body of that component.

A midsurface feature is calculated from the solid body in your part document. This method first finds all parallel pairs of faces from the part solid. For each pair of parallel faces found, it creates a neutral reference surface between the two faces. The placement position of the reference surface is controlled by the placement parameter. This process repeats itself until all parallel body faces are processed. At the end of this process, all the reference surfaces are sewn into a single reference surface if knitFlag is set to true.

The hierarchy of a midsurface feature is a midsurface feature contains one or more reference surfaces. If sewn successfully using the KnitFlag = True option, then the midsurface feature contains only one reference surface. Each reference surface is considered a sheet body. The sheet body has the normal topology that you would expect to find on a body object (for example, faces, edges, and so on). See [IMidSurface3](#) for methods that provide access to this topology.

See Also

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMirrorFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMirrorFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMirrorFeature2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMirrorFeature( _  
    ByVal BMirrorBody As System.Boolean, _  
    ByVal BGeometryPattern As System.Boolean, _  
    ByVal BMerge As System.Boolean, _  
    ByVal BKnit As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BMirrorBody As System.Boolean  
Dim BGeometryPattern As System.Boolean  
Dim BMerge As System.Boolean  
Dim BKnit As System.Boolean  
Dim value As Feature  
  
value = instance.InsertMirrorFeature(BMirrorBody, BGeometryPattern, BMerge, BKnit)
```

C#

```
Feature InsertMirrorFeature(  
    System.bool BMirrorBody,  
    System.bool BGeometryPattern,  
    System.bool BMerge,  
    System.bool BKnit  
)
```

C++/CLI

```
Feature^ InsertMirrorFeature(  
& System.bool BMirrorBody,  
& System.bool BGeometryPattern,  
& System.bool BMerge,  
& System.bool BKnit  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BMirrorBody

True to mirror solid bodies; false to mirror a feature or face

BGeometryPattern

True to mirror only the feature geometry, false to solve the entire feature; applies to mirroring features only

BMerge

True to merge any mirrored solid bodies, false to not; applies to mirroring solid bodies only

BKnit

True to knit surfaces, false to not; applies to mirroring surfaces only

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMirrorFeature](#).

Remarks

This method attempts to create the mirror feature without displaying a dialog box to get information from the user. It relies on preselected and marked entities, as well as arguments.

Any...	Must be preselected and marked with a value of...
Features to be mirrored	1
Faces to be mirrored	128
Bodies to be mirrored	256
Plane or planar face	2

For information on selecting and marking entities, see [IModelDocExtension::SelectById2](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMirrorSolidFeatureData Interface](#)
[IMirrorPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMirrorFeature2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMirrorFeature2 Method (IFeatureManager)

Mirrors selected features, faces, bodies, and structure systems about a selected plane or planar face.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMirrorFeature2( _  
    ByVal BMirrorBody As System.Boolean, _  
    ByVal BGeometryPattern As System.Boolean, _  
    ByVal BMerge As System.Boolean, _  
    ByVal BKnit As System.Boolean, _  
    ByVal ScopeOptions As System.Integer _  
) As Feature
```

Visual Basic (Usage) 

```
Dim instance As IFeatureManager  
Dim BMirrorBody As System.Boolean  
Dim BGeometryPattern As System.Boolean  
Dim BMerge As System.Boolean  
Dim BKnit As System.Boolean  
Dim ScopeOptions As System.Integer  
Dim value As Feature  
  
value = instance.InsertMirrorFeature2(BMirrorBody, BGeometryPattern, BMerge, BKnit, ScopeOptions)
```

C#

```
Feature InsertMirrorFeature2(  
    System.bool BMirrorBody,  
    System.bool BGeometryPattern,  
    System.bool BMerge,  
    System.bool BKnit,  
    System.int ScopeOptions  
)
```

C++/CLI

```
Feature^ InsertMirrorFeature2(  
& System.bool BMirrorBody,  
& System.bool BGeometryPattern,  
& System.bool BMerge,  
& System.bool BKnit,  
& System.int ScopeOptions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BMirrorBody
True to mirror solid bodies; false to mirror a feature or face

BGeometryPattern
True to mirror only the feature geometry, false to solve the entire feature; applies to mirroring features only

BMerge
True to merge any mirrored solid bodies, false to not; applies to mirroring solid bodies only

BKnit
True to knit surfaces, false to not; applies to mirroring surfaces only

ScopeOptions
Feature scope as defined by [swFeatureScope_e](#)

Return Value

[Feature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMirrorFeature2](#).

 Example

See the [IMirrorSolidFeatureData::StructureSystemToPatternArray](#) examples.

 Example

[Get Mirror Pattern Feature Data \(C#\)](#)
[Get Mirror Pattern Feature Data \(VB.NET\)](#)
[Get Mirror Pattern Feature Data \(VBA\)](#)

 Remarks

This method attempts to create the mirror feature without displaying a dialog box to get information from the user. It relies on preselected and marked entities and arguments.

Structure systems 134217728

Any of these entities to be mirrored...	Must be preselected and marked with a value of...
Features	1
Faces	128
Bodies	256
Structure systems	134217728
Planes or planar faces	2

For information on selecting and marking entities, see [IModelDocExtension::SelectById2](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMirrorPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoldCoreCavitySolids Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoldCoreCavitySolids Method (IFeatureManager)

Creates a core/cavity solid feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoldCoreCavitySolids( _  
    ByVal Dist1 As System.Double, _  
    ByVal Dist2 As System.Double, _  
    ByVal Setback As System.Boolean, _  
    ByVal Angle As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Dist1 As System.Double  
Dim Dist2 As System.Double  
Dim Setback As System.Boolean  
Dim Angle As System.Double  
Dim value As Feature  
  
value = instance.InsertMoldCoreCavitySolids(Dist1, Dist2, Setback, Angle)
```

C#

```
Feature InsertMoldCoreCavitySolids(  
    System.double Dist1,  
    System.double Dist2,  
    System.bool Setback,  
    System.double Angle  
)
```

C++/CLI

```
Feature^ InsertMoldCoreCavitySolids(  
& System.double Dist1,  
& System.double Dist2,  
& System.bool Setback,  
& System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Dist1

Length of the extrusion in Direction 1

Dist2

Length of the extrusion in Direction 2

Setback

True to enable setback surface, false to not

Angle

Draft angle for the setback surface

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoldCoreCavitySolids](#).

Remarks

To use this method:

- The Parting Line feature must exist.
- The parting surface must be generated.
- The part must be separated into core and cavity (sheet bodies).
- A sketch describing the outline of the mold block must exist. The sketch must be perpendicular to the pull direction. Usually, the sketch is a rectangle that includes the part (core and cavity).

The location of the sketch plane specifies the parting plane in the setback surface option.

- Without the setback surface option, the parting surface must extend beyond the block sketch so that the parting surface is cut off by the block.
- With the setback surface option, the parting surface should be included within the block sketch. The parting plane (sketch plane) cannot intersect with the parting surface. Therefore, the parting plane must be either higher or lower than the parting surface. From the parting surface a set of ruled surfaces is generated to be trimmed by the parting plane.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertMoldPartingLine Method](#)
[IFeatureManager.InsertMoldPartingSurface Method](#)
[IFeatureManager.InsertMoldShutOffSurface Method](#)
[ICoreFeatureData Interface](#)
[ICavityFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoldPartingLine Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoldPartingLine Method (IFeatureManager)

Inserts a mold parting line feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoldPartingLine( _  
    ByVal BFlipDir As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BFlipDir As System.Boolean  
Dim value As Feature
```

```
value = instance.InsertMoldPartingLine(BFlipDir)
```

C#

```
Feature InsertMoldPartingLine(  
    System.bool BFlipDir  
)
```

C++/CLI

```
Feature^ InsertMoldPartingLine(  
&    System.bool BFlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BFlipDir
True to flip the direction of the pull, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMoldPartingLine](#).

Remarks

Before calling this method, you must select the direction of pull and the edges for the parting line by calling [IModelDocExtension::SelectByID2](#) with the following Marks:

Selection	Mark
Direction of pull	1
Parting line edges	4

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IPartingLineFeatureData Interface](#)
[IFeatureManager::InsertMoldCoreCavitySolids Method](#)
[IFeatureManager::InsertMoldShutOffSurface Method](#)
[IFeatureManager::InsertMoldCoreCavitySolids Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoldPartingSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoldPartingSurface Method (IFeatureManager)

Inserts a mold parting surface feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoldPartingSurface( _  
    ByVal Radiate As System.Integer, _  
    ByVal ReverseAlignment As System.Boolean, _  
    ByVal ReverseOffset As System.Boolean, _  
    ByVal OffsetDistance As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal Smooth As System.Integer, _  
    ByVal SmoothDistance As System.Double, _  
    ByVal Knit As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Radiate As System.Integer  
Dim ReverseAlignment As System.Boolean  
Dim ReverseOffset As System.Boolean  
Dim OffsetDistance As System.Double  
Dim Angle As System.Double  
Dim Smooth As System.Integer  
Dim SmoothDistance As System.Double  
Dim Knit As System.Boolean  
Dim value As Feature  
  
value = instance.InsertMoldPartingSurface(Radiate, ReverseAlignment, ReverseOffset, OffsetDistance, Angle, Smooth, SmoothDistance, Knit)
```

C#

```
Feature InsertMoldPartingSurface(  
    System.int Radiate,  
    System.bool ReverseAlignment,  
    System.bool ReverseOffset,  
    System.double OffsetDistance,  
    System.double Angle,  
    System.int Smooth,  
    System.double SmoothDistance,  
    System.bool Knit  
)
```

C++/CLI

```
Feature^ InsertMoldPartingSurface(  
& System.int Radiate,  
& System.bool ReverseAlignment,  
& System.bool ReverseOffset,  
& System.double OffsetDistance,  
& System.double Angle,  
& System.int Smooth,  
& System.double SmoothDistance,  
& System.bool Knit  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Radiate

Radiate mold parting surface as defined by [swPartingSurfaceMoldParmType_e](#)

ReverseAlignment

True to reverse alignment, false to not; only available when radiate set to swPartingSurfaceMoldParmNormal and a parting line does not yet exist (see **Remarks**)

ReverseOffset

True to reverse offset direction, false to not

OffsetDistance

True to reverse offset direction, false to not

Angle

Angle of mold parting surface; only available when radiate set to either swPartingSurfaceMoldParmTangent or swPartingSurfaceMoldParmNormal

Smooth

Smooth mold parting surface as defined by [swPartingSurfaceSmoothingType_e](#)

SmoothDistance

Distance to smooth mold parting surface; only available when smooth set to swPartingSurfaceSmooth

Knit

True to knit all surfaces, false to not

Return Value

Pointer to [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMoldPartingSurface](#).

Remarks

If a parting line feature does not yet exist in the model, you must first select the direction of pull and the edges for the parting line using [IModelDocExtension::SelectById2](#). For example, a face (direction of pull) has a mark of 1 and edges (parting lines) a mark of 4.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IPartingSurfaceFeatureData Interface](#)

[IFeatureManager::InsertMoldCoreCavitySolids Method](#)

[IFeatureManager::InsertMoldPartingLine Method](#)

[IFeatureManager::InsertMoldShutOffSurface Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoldShutOffSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoldShutOffSurface Method (IFeatureManager)

Inserts the mold shut-off surface feature.

 .NET Syntax

Visual Basic (Declaration)

Function InsertMoldShutOffSurface() As Feature

Visual Basic (Usage)

Copy Code

Dim instance As [IFeatureManager](#)
Dim value As Feature

value = instance.InsertMoldShutOffSurface()

C#

Feature InsertMoldShutOffSurface()

C++/CLI

Feature^ InsertMoldShutOffSurface();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoldShutOffSurface](#).

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IShutOffSurfaceFeatureData Interface](#)
[IFeatureManager.InsertMoldCoreCavitySolids Method](#)
[IFeatureManager.InsertMoldPartingLine Method](#)
[IFeatureManager.InsertMoldPartingSurface Method](#)

 Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help [Send comments on this topic.](#)

InsertMoveCopyBody Method (IFeatureManager)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveCopyBody Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMoveCopyBody2](#).

 .NET Syntax

Visual Basic (Declaration)

Function InsertMoveCopyBody(_
ByVal [TransX](#) As System.Double, _
ByVal [TransY](#) As System.Double, _
ByVal [TransZ](#) As System.Double, _
ByVal [TransDist](#) As System.Double, _
ByVal [RotPointX](#) As System.Double, _
ByVal [RotPointY](#) As System.Double, _
ByVal [RotPointZ](#) As System.Double, _
ByVal [RotAngleX](#) As System.Double, _
ByVal [RotAngleY](#) As System.Double, _
ByVal [RotAngleZ](#) As System.Double, _
ByVal [BCopy](#) As System.Boolean, _
ByVal [NumCopies](#) As System.Integer _
) As Feature

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim TransX As System.Double
Dim TransY As System.Double
Dim TransZ As System.Double
Dim TransDist As System.Double
Dim RotPointX As System.Double
Dim RotPointY As System.Double
Dim RotPointZ As System.Double
Dim RotAngleX As System.Double
Dim RotAngleY As System.Double
Dim RotAngleZ As System.Double
Dim BCopy As System.Boolean
Dim NumCopies As System.Integer
Dim value As Feature
```

```
value = instance.InsertMoveCopyBody(TransX, TransY, TransZ, TransDist, RotPointX, RotPointY, RotPointZ,
RotAngleX, RotAngleY, RotAngleZ, BCopy, NumCopies)
```

C#

```
Feature InsertMoveCopyBody(
    System.double TransX,
    System.double TransY,
    System.double TransZ,
    System.double TransDist,
    System.double RotPointX,
    System.double RotPointY,
    System.double RotPointZ,
    System.double RotAngleX,
    System.double RotAngleY,
    System.double RotAngleZ,
    System.bool BCopy,
    System.int NumCopies
)
```

C++/CLI

```
Feature^ InsertMoveCopyBody(
    & System.double TransX,
    & System.double TransY,
    & System.double TransZ,
    & System.double TransDist,
    & System.double RotPointX,
    & System.double RotPointY,
    & System.double RotPointZ,
    & System.double RotAngleX,
    & System.double RotAngleY,
    & System.double RotAngleZ,
    & System.bool BCopy,
    & System.int NumCopies
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransX

TransY

TransZ

TransDist

RotPointX

RotPointY

RotPointZ

RotAngleX

RotAngleY

RotAngleZ

BCopy

NumCopies

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoveCopyBody](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoveCopyBody2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveCopyBody2 Method (IFeatureManager)

Moves or makes copies of the selected solid bodies or surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoveCopyBody2( _  
    ByVal TransX As System.Double, _  
    ByVal TransY As System.Double, _  
    ByVal TransZ As System.Double, _  
    ByVal TransDist As System.Double, _  
    ByVal RotPointX As System.Double, _  
    ByVal RotPointY As System.Double, _  
    ByVal RotPointZ As System.Double, _  
    ByVal RotAngleX As System.Double, _  
    ByVal RotAngleY As System.Double, _  
    ByVal RotAngleZ As System.Double, _  
    ByVal BCopy As System.Boolean, _  
    ByVal NumCopies As System.Integer _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim TransX As System.Double  
Dim TransY As System.Double  
Dim TransZ As System.Double  
Dim TransDist As System.Double  
Dim RotPointX As System.Double  
Dim RotPointY As System.Double  
Dim RotPointZ As System.Double  
Dim RotAngleX As System.Double  
Dim RotAngleY As System.Double  
Dim RotAngleZ As System.Double  
Dim BCopy As System.Boolean  
Dim NumCopies As System.Integer  
Dim value As Feature
```

```
value = instance.InsertMoveCopyBody2(TransX, TransY, TransZ, TransDist, RotPointX, RotPointY,  
RotPointZ, RotAngleX, RotAngleY, RotAngleZ, BCopy, NumCopies)
```

C#

```
Feature InsertMoveCopyBody2(  
    System.double TransX,  
    System.double TransY,  
    System.double TransZ,  
    System.double TransDist,  
    System.double RotPointX,  
    System.double RotPointY,  
    System.double RotPointZ,  
    System.double RotAngleX,  
    System.double RotAngleY,  
    System.double RotAngleZ,  
    System.bool BCopy,
```

```
System.int NumCopies  
)
```

C++/CLI

```
Feature^ InsertMoveCopyBody2(  
& System.double TransX,  
& System.double TransY,  
& System.double TransZ,  
& System.double TransDist,  
& System.double RotPointX,  
& System.double RotPointY,  
& System.double RotPointZ,  
& System.double RotAngleX,  
& System.double RotAngleY,  
& System.double RotAngleZ,  
& System.bool BCopy,  
& System.int NumCopies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransX
Value for delta X; applies to translation and valid only if Rot* parameters are null or Nothing (see **Remarks**)
TransY
Value for delta Y; applies to translation and valid only if Rot* parameters are null or Nothing (see **Remarks**)
TransZ
Value for delta Z; applies to translation and valid only if Rot* parameters are null or Nothing (see **Remarks**)
TransDist
Distance ; applies to translation and valid only if Rot* parameters are null or Nothing (see **Remarks**)
RotPointX
Value for X rotation origin; applies to rotation (see **Remarks**)
RotPointY
Value for Y rotation origin; applies to rotation (see **Remarks**)
RotPointZ
Value for Z rotation origin; applies to rotation (see **Remarks**)
RotAngleX
Value for X rotation angle; applies to rotation (see **Remarks**)
RotAngleY
Value for Y rotation angle; applies to rotation (see **Remarks**)
RotAngleZ
Value for Z rotation angle; applies to rotation (see **Remarks**)
BCopy
True if a copy operation, false if a move operation
NumCopies
Number of copies to create

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoveCopyBody2](#).

Example

[Move Bodies \(C#\)](#)
[Move Bodies \(VB.NET\)](#)
[Move Bodies \(VBA\)](#)

Remarks

Specify either translation or rotation parameters. Translation and rotation cannot both be applied by a single move/copy body feature. If you specify both translation and rotation parameters, only the rotation parameters are applied.

This method requires selecting the body, surface, edge, and vertex using these marks:

Selection	Mark
Body or surface	1
Rotation edge, vertex, coordinate system	2
Translation edge, vertex, coordinate system	4
Translation vertex	8

NOTE: To add the move/copy body as a sub-feature of a derived part, you must select the move/copy body feature first.

See [IModelDocExtension::SelectByID2](#) for details on selecting and marking bodies, surfaces, edges, vertices, and coordinate systems.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMoveCopyBodyFeatureData Interface](#)

Availability

SOLIDWORKS 2003 SP3, Revision Number 11.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoveFace Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveFace Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertMoveFace2](#) and [IFeatureManager::InsertMoveFace2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoveFace( _  
    ByVal MoveType As System.Integer, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal Distance As System.Double _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim MoveType As System.Integer  
Dim ReverseDir As System.Boolean  
Dim Angle As System.Double  
Dim Distance As System.Double  
Dim value As Feature  
  
value = instance.InsertMoveFace(MoveType, ReverseDir, Angle, Distance)
```

C#

```
Feature InsertMoveFace(  
    System.int MoveType,  
    System.bool ReverseDir,  
    System.double Angle,  
    System.double Distance  
)
```

C++/CLI

```
Feature^ InsertMoveFace(  
& System.int MoveType,  
& System.bool ReverseDir,  
& System.double Angle,  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MoveType

Type of move:

- 0 = Offset
- 1 = Translate
- 2 = Rotate

ReverseDir

True to reverse the direction, false to not

Angle

If MoveType is Rotate, then specify the angle at which to draft the faces

Distance

Distance to offset or translate the faces

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoveFace](#).

Example

[Move Selected Face \(VBA\)](#)

Remarks

Use the following marks with [IModelDocExtension::SelectByID2](#):

- 1 = face
- 2 = direction reference (plane, planar face, linear edge, or reference axis) for translate
- 4 = axis reference (linear edge or reference axis) for rotate

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IMoveFaceFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoveFace2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveFace2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMoveFace3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMoveFace2( _  
    ByVal MoveType As System.Integer, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal Distance As System.Double, _  
    ByVal TranslationParams As System.Object, _  
    ByVal RotationParams As System.Object _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim MoveType As System.Integer  
Dim ReverseDir As System.Boolean  
Dim Angle As System.Double  
Dim Distance As System.Double  
Dim TranslationParams As System.Object  
Dim RotationParams As System.Object  
Dim value As Feature  
  
value = instance.InsertMoveFace2(MoveType, ReverseDir, Angle, Distance, TranslationParams,  
    RotationParams)
```

C#

```
Feature InsertMoveFace2(  
    System.int MoveType,  
    System.bool ReverseDir,
```

```

System.double Angle,
System.double Distance,
System.object TranslationParams,
System.object RotationParams
)

```

C++/CLI

```

Feature^ InsertMoveFace2(
& System.int MoveType,
& System.bool ReverseDir,
& System.double Angle,
& System.double Distance,
& System.Object^ TranslationParams,
& System.Object^ RotationParams
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MoveType

Type of move:

- 0 = Offset
- 1 = Translate
- 2 = Rotate

ReverseDir

True to reverse the direction, false to not

Angle

If MoveType is Rotate, then specify the angle at which to draft the faces

Distance

Distance to offset or translate the faces

TranslationParams

Array of three doubles for the delta x, delta y, and delta z direction translation

RotationParams

Array of six doubles:

- First three doubles are the x, y, and z rotation origin
- Last three doubles are the x, y, and z rotation angle

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMoveFace2](#).

Remarks

Use the following marks with [IModelDocExtension::SelectByID2](#):

- 1 = face
- 2 = direction reference (plane, planar face, linear edge, or reference axis) for translate
- 4 = axis reference (linear edge or reference axis) for rotate

If you specify a value for TranslationParams or RotationParams, then do not specify a value for Distance or Angle, respectively. The translation or rotation parameters are calculated internally when Distance or Angle is specified.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::InsertMoveFace2 Method](#)

[IMoveFaceFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoveFace3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveFace3 Method (IFeatureManager)

Moves the selected faces on a solid or surface model.

Visual Basic (Declaration)

```
Function InsertMoveFace3( _
    ByVal MoveType As System.Integer, _
    ByVal ReverseDir As System.Boolean, _
    ByVal Angle As System.Double, _
    ByVal Distance As System.Double, _
    ByVal TranslationParams As System.Object, _
    ByVal RotationParams As System.Object, _
    ByVal EndConditionType As System.Integer, _
    ByVal OffsetDistance As System.Double _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim MoveType As System.Integer
Dim ReverseDir As System.Boolean
Dim Angle As System.Double
Dim Distance As System.Double
Dim TranslationParams As System.Object
Dim RotationParams As System.Object
Dim EndConditionType As System.Integer
Dim OffsetDistance As System.Double
Dim value As Feature

value = instance.InsertMoveFace3(MoveType, ReverseDir, Angle, Distance, TranslationParams,
RotationParams, EndConditionType, OffsetDistance)
```

C#

```
Feature InsertMoveFace3(
    System.int MoveType,
    System.bool ReverseDir,
    System.double Angle,
    System.double Distance,
    System.object TranslationParams,
    System.object RotationParams,
    System.int EndConditionType,
    System.double OffsetDistance
)
```

C++/CLI

```
Feature^ InsertMoveFace3(
    & System.int MoveType,
    & System.bool ReverseDir,
    & System.double Angle,
    & System.double Distance,
    & System.Object^ TranslationParams,
    & System.Object^ RotationParams,
    & System.int EndConditionType,
    & System.double OffsetDistance
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MoveType

Type of move as defined in [swMoveFaceType_e](#)

ReverseDir

True to reverse the direction, false to not

Angle

Angle at which to draft the faces; valid only if MoveType is swMoveFaceType_e.swMoveFaceTypeRotate (see **Remarks**)

Distance

Distance to translate or offset the faces; valid only if MoveType is one of the following:

- swMoveFaceType_e.swMoveFaceTypeOffset
- swMoveFaceType_e.swMoveFaceTypeTranslate and EndConditionType is swEndConditions_e.swEndCondBlind (see **Remarks**)

TranslationParams

Array of three doubles for the delta x, delta y, and delta z direction translation (see **Remarks**)

RotationParams

Array of six doubles:

- First three doubles are the x, y, and z rotation origin
- Last three doubles are the x, y, and z rotation angle

(see **Remarks**)
EndConditionType
End condition as defined in [swEndConditions_e](#); valid only if MoveType is swMoveFaceType_e.swMoveFaceTypeTranslate

Only the following end condition types are valid:

- swEndConditions_e.swEndCondBlind
- swEndConditions_e.swEndCondUpToVertex
- swEndConditions_e.swEndCondUpToSurface
- swEndConditions_e.swEndCondOffsetFromSurface
- swEndConditions_e.swEndCondUpToBody

OffsetDistance
Offset from surface; valid only if MoveType is swMoveFaceType_e.swMoveFaceTypeTranslate and EndConditionType is swEndConditions_e.swEndCondOffsetFromSurface

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMoveFace3](#).

Example

[Create and Modify Move Face Feature \(VBA\)](#)
[Create and Modify Move Face Feature \(VB.NET\)](#)
[Create and Modify Move Face Feature \(C#\)](#)

Remarks

Before calling this method, you need to select specific entities.

If MoveFaceType is swMoveFaceType_e...	And EndConditionType is swEndConditions_e...	Call IModelDocExtension::SelectByID2 to select...	With mark...
Any option	Any option	Face to move	1
swMoveFaceTypeTranslate	Any option	Direction reference (plane, planar face, linear edge, or reference axis)	2
swMoveFaceTypeTranslate	• swEndCondUpToVertex • swEndCondUpToSurface • swEndCondOffsetFromSurface • swEndCondUpToBody	• Up-to vertex • Up-to surface • Offset-from surface • Up-to body	8
swMoveFaceTypeRotate	N/A	Axis reference (linear edge or reference axis)	4

If you specify a value for TranslationParams or RotationParams, then do not specify a value for Distance or Angle, respectively. The translation or rotation parameters are calculated internally when Distance or Angle is specified.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMoveFaceFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMultiFaceDraft Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMultiFaceDraft Method (IFeatureManager)

Inserts a multiface draft feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMultiFaceDraft( _  
    ByVal Angle As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal EdgeDraft As System.Boolean, _  
    ByVal PropType As System.Integer, _  
    ByVal IsStepDraft As System.Boolean, _  
    ByVal IsBodyDraft As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

Copy Code

```

Dim instance As IFeatureManager
Dim Angle As System.Double
Dim FlipDir As System.Boolean
Dim EdgeDraft As System.Boolean
Dim PropType As System.Integer
Dim IsStepDraft As System.Boolean
Dim IsBodyDraft As System.Boolean
Dim value As Feature

```

```
value = instance.InsertMultiFaceDraft(Angle, FlipDir, EdgeDraft, PropType, IsStepDraft, IsBodyDraft)
```

C#

```

Feature InsertMultiFaceDraft(
    System.double Angle,
    System.bool FlipDir,
    System.bool EdgeDraft,
    System.int PropType,
    System.bool IsStepDraft,
    System.bool IsBodyDraft
)

```

C++/CLI

```

Feature^ InsertMultiFaceDraft(
& System.double Angle,
& System.bool FlipDir,
& System.bool EdgeDraft,
& System.int PropType,
& System.bool IsStepDraft,
& System.bool IsBodyDraft
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Draft angle in radians

FlipDir

True to flip the direction of the draft (the draft direction is determined by the selected direction, which could be a planar face, a reference plane, an edge, or two vertices; if you refer to the last argument in [IModelDocExtension::SelectById2](#), then these items have a selection mark of 1; the faces to draft have a selection mark of 2, and the draft edges have a selection mark of 4), false to not

EdgeDraft

True if this is an edge draft and the user selects edges to draft, false if this is a face draft and the user selects faces to draft (SOLIDWORKS automatically selects the faces next to the edge to be drafted)

PropType

Propagation type:

- 0 = no propagation
- 1 = propagate to the next tangent faces that share the base neutral plane; valid for edge draft or neutral plane draft
- 2 = propagate to all faces that are a neighbor of the neutral plane
- 3 = propagate to all faces that are a neighbor of the neutral plane on all inner loops (for example, all faces of a pocket or a boss)
- 4 = propagate to all faces that are a neighbor of the neutral plane on the outer loop (for example, the sides of a box with the bottom face as the neutral plane)

IsStepDraft

True if the draft is a step draft, false if not

IsBodyDraft

True if the draft is a body draft, false if not

Return Value

Pointer to the newly created draft [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertMultiFaceDraft](#).

Remarks

Make the selections using [IModelDocExtension::SelectById2](#) before calling this method.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IDraftFeatureData2 Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertNetBlend Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertNetBlend Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertNetBlend2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNetBlend( _  
    ByVal Type As System.Short, _  
    ByVal NCurvesDir1 As System.Short, _  
    ByVal NCurvesDir2 As System.Short, _  
    ByVal HasCenterline As System.Boolean, _  
    ByVal TessTolFactor As System.Double, _  
    ByVal WantsSolid As System.Boolean, _  
    ByVal MergeBody As System.Boolean, _  
    ByVal FeatureScope As System.Boolean, _  
    ByVal AutoSelect As System.Boolean, _  
    ByVal Thin As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ReverseThickness As System.Boolean, _  
    ByVal ThinSolidType As System.Short, _  
    ByVal UseSharedThickness As System.Boolean, _  
    ByVal CapEnds As System.Boolean, _  
    ByVal EndThickness As System.Double, _  
    ByVal AutoFillet As System.Boolean, _  
    ByVal FilletRadius As System.Double, _  
    ByVal ForceNonRational As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Short  
Dim NCurvesDir1 As System.Short  
Dim NCurvesDir2 As System.Short  
Dim HasCenterline As System.Boolean  
Dim TessTolFactor As System.Double  
Dim WantsSolid As System.Boolean  
Dim MergeBody As System.Boolean  
Dim FeatureScope As System.Boolean  
Dim AutoSelect As System.Boolean  
Dim Thin As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ReverseThickness As System.Boolean  
Dim ThinSolidType As System.Short  
Dim UseSharedThickness As System.Boolean  
Dim CapEnds As System.Boolean  
Dim EndThickness As System.Double  
Dim AutoFillet As System.Boolean  
Dim FilletRadius As System.Double  
Dim ForceNonRational As System.Boolean  
Dim value As Feature  
  
value = instance.InsertNetBlend(Type, NCurvesDir1, NCurvesDir2, HasCenterline, TessTolFactor,  
WantsSolid, MergeBody, FeatureScope, AutoSelect, Thin, Thickness1, Thickness2, ReverseThickness,  
ThinSolidType, UseSharedThickness, CapEnds, EndThickness, AutoFillet, FilletRadius, ForceNonRational)
```

C#

```
Feature InsertNetBlend(  
    System.short Type,
```

```

System.short NCurvesDir1,
System.short NCurvesDir2,
System.bool HasCenterline,
System.double TessTolFactor,
System.bool WantsSolid,
System.bool MergeBody,
System.bool FeatureScope,
System.bool AutoSelect,
System.bool Thin,
System.double Thickness1,
System.double Thickness2,
System.bool ReverseThickness,
System.short ThinSolidType,
System.bool UseSharedThickness,
System.bool CapEnds,
System.double EndThickness,
System.bool AutoFillet,
System.double FilletRadius,
System.bool ForceNonRational
)

```

C++/CLI

```

Feature^ InsertNetBlend(
& System.short Type,
& System.short NCurvesDir1,
& System.short NCurvesDir2,
& System.bool HasCenterline,
& System.double TessTolFactor,
& System.bool WantsSolid,
& System.bool MergeBody,
& System.bool FeatureScope,
& System.bool AutoSelect,
& System.bool Thin,
& System.double Thickness1,
& System.double Thickness2,
& System.bool ReverseThickness,
& System.short ThinSolidType,
& System.bool UseSharedThickness,
& System.bool CapEnds,
& System.double EndThickness,
& System.bool AutoFillet,
& System.double FilletRadius,
& System.bool ForceNonRational
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

- 0 = Base
- 1 = Boss
- 2 = Surface
- 3 = Cut

NCurvesDir1

Number of curves in Direction 1

NCurvesDir2

Number of curves in Direction 2

HasCenterline

True if a centerline exists, false if not

TessTolFactor

Factor to control the number of intermediate sections used for loft with centerline; default value is 1.0; the greater the value, the more intermediate sections created

WantsSolid

True to create a solid body, false to create a surface body; only valid when Type = 0, 1, or 3

MergeBody

True merges the results in a multibody part, false does not

FeatureScope

True if the feature only affects selected bodies, false if the feature affects all bodies

AutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

Thin

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ReverseThickness

True to reverse the thickness, false to not

ThinSolidType

Thin wall type as defined in [swThinWallType_e](#)

UseSharedThickness

True to use shared thickness, false to not

CapEnds

True to cap the ends, false to not

EndThickness

End cap thickness
AutoFillet
True to fillet the corners automatically, false to not
FilletRadius
Fillet radius when AutoFillet is true
ForceNonRational
True to force the resulting surface to be non-rational, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertNetBlend](#).

Example

[Insert Boundary Surface Feature \(VBA\)](#)
[Insert Boundary Feature \(C#\)](#)
[Insert Boundary Feature \(VB.NET\)](#)
[Insert Boundary Feature \(VBA\)](#)

Remarks

Boundary feature

Use [IModelDocExtension::SelectByID2](#) to select the entities for a boundary feature or boundary surface feature. The following table contains valid selection marks for this method and these features.

Feature	Selection marks
Boundary	Sketch curves: - Direction 1: 8193, 16385, 24577 - Direction 2: 8194, 16386, 24578 Faces: - Direction 1: 8193, 16385, 24577 - Direction 2: 8194, 16386, 24578 Other sketch entities (3D sketch (Unknown/SELOBJGROUP): - Direction 1: 12289, 24577 - Direction 2: 16386, 28674
Boundary surface	Sketch curves: - Direction 1: 1 - Direction 2: 2

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::SetNetBlendCurveData Method](#)
[IFeatureManager::SetNetBlendDirectionData Method](#)
[IBoundaryBossFeatureData Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

InsertNetBlend2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertNetBlend2 Method (IFeatureManager)

This method inserts a boundary feature or a boundary surface feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNetBlend2( _  
    ByVal Type As System.Short, _  
    ByVal NCurvesDir1 As System.Short, _  
    ByVal NCurvesDir2 As System.Short, _  
    ByVal HasCenterline As System.Boolean, _  
    ByVal TessTolFactor As System.Double, _  
    ByVal WantsSolid As System.Boolean, _  
    ByVal MergeBody As System.Boolean, _
```

```

ByVal FeatureScope As System.Boolean, _
ByVal AutoSelect As System.Boolean, _
ByVal Thin As System.Boolean, _
ByVal Thickness1 As System.Double, _
ByVal Thickness2 As System.Double, _
ByVal ReverseThickness As System.Boolean, _
ByVal ThinSolidType As System.Short, _
ByVal UseSharedThickness As System.Boolean, _
ByVal CapEnds As System.Boolean, _
ByVal EndThickness As System.Double, _
ByVal AutoFillet As System.Boolean, _
ByVal FilletRadius As System.Double, _
ByVal ForceNonRational As System.Boolean, _
ByVal CreateSolid As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Type As System.Short
Dim NCurvesDir1 As System.Short
Dim NCurvesDir2 As System.Short
Dim HasCenterline As System.Boolean
Dim TessTolFactor As System.Double
Dim WantsSolid As System.Boolean
Dim MergeBody As System.Boolean
Dim FeatureScope As System.Boolean
Dim AutoSelect As System.Boolean
Dim Thin As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ReverseThickness As System.Boolean
Dim ThinSolidType As System.Short
Dim UseSharedThickness As System.Boolean
Dim CapEnds As System.Boolean
Dim EndThickness As System.Double
Dim AutoFillet As System.Boolean
Dim FilletRadius As System.Double
Dim ForceNonRational As System.Boolean
Dim CreateSolid As System.Boolean
Dim value As Feature

```

```

value = instance.InsertNetBlend2(Type, NCurvesDir1, NCurvesDir2, HasCenterline, TessTolFactor,
WantsSolid, MergeBody, FeatureScope, AutoSelect, Thin, Thickness1, Thickness2, ReverseThickness,
ThinSolidType, UseSharedThickness, CapEnds, EndThickness, AutoFillet, FilletRadius, ForceNonRational,
CreateSolid)

```

C#

```

Feature InsertNetBlend2(
    System.short Type,
    System.short NCurvesDir1,
    System.short NCurvesDir2,
    System.bool HasCenterline,
    System.double TessTolFactor,
    System.bool WantsSolid,
    System.bool MergeBody,
    System.bool FeatureScope,
    System.bool AutoSelect,
    System.bool Thin,
    System.double Thickness1,
    System.double Thickness2,
    System.bool ReverseThickness,
    System.short ThinSolidType,
    System.bool UseSharedThickness,
    System.bool CapEnds,
    System.double EndThickness,
    System.bool AutoFillet,
    System.double FilletRadius,
    System.bool ForceNonRational,
    System.bool CreateSolid
)

```

```

Feature^ InsertNetBlend2(
&    System.short Type,
&    System.short NCurvesDir1,
&    System.short NCurvesDir2,
&    System.bool HasCenterline,
&    System.double TessTolFactor,
&    System.bool WantsSolid,
&    System.bool MergeBody,
&    System.bool FeatureScope,
&    System.bool AutoSelect,
&    System.bool Thin,
&    System.double Thickness1,
&    System.double Thickness2,
&    System.bool ReverseThickness,
&    System.short ThinSolidType,
&    System.bool UseSharedThickness,
&    System.bool CapEnds,
&    System.double EndThickness,
&    System.bool AutoFillet,
&    System.double FilletRadius,
&    System.bool ForceNonRational,
&    System.bool CreateSolid
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

- 0 = Base
- 1 = Boss
- 2 = Surface
- 3 = Cut

NCurvesDir1

Number of curves in Direction 1

NCurvesDir2

Number of curves in Direction 2

HasCenterline

True if a centerline exists, false if not

TessTolFactor

Factor to control the number of intermediate sections used for loft with centerline; default value is 1.0; the greater the value, the more intermediate sections created

WantsSolid

True to create a solid body, false to create a surface body; only valid when Type = 0, 1, or 3

MergeBody

True merges the results in a multibody part, false does not

FeatureScope

True if the feature only affects selected bodies, false if the feature affects all bodies

AutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

Thin

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ReverseThickness

True to reverse the thickness, false to not

ThinSolidType

Thin wall type as defined in [swThinWallType_e](#)

UseSharedThickness

True to use shared thickness, false to not

CapEnds

True to cap the ends, false to not

EndThickness

End cap thickness

AutoFillet

True to fillet the corners automatically, false to not

FilletRadius

Fillet radius when AutoFillet is true

ForceNonRational

True to force the resulting surface to be non-rational, false to not

CreateSolid

True to create a solid body, false to create a surface body; only valid when Type = 2

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertNetBlend2](#).

Example

[Insert Solid Body Boundary Surface Feature \(C#\)](#)

[Insert Solid Body Boundary Surface Feature \(VB.NET\)](#)

[Insert Solid Body Boundary Surface Feature \(VBA\)](#)

Remarks

Use [IModelDocExtension::SelectByID2](#) and the following selection marks to select the entities for a boundary feature or boundary surface feature.

- Sketch curves
 - Direction 1: 8193, 16385, 24577
 - Direction 2: 8194, 16386, 24578
- Faces
 - Direction 1: 8193, 16385, 24577
 - Direction 2: 8194, 16386, 24578
- Other sketch entities (3D sketch (Unknown/SELOBJGROUP))
 - Direction 1: 12289, 24577
 - Direction 2: 16386, 28674

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::SetNetBlendCurveData Method \(\)](#)
[IFeatureManager::SetNetBlendDirectionData Method \(\)](#)
[IBoundaryBossFeatureData Interface](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionBlend Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertProtrusionBlend Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionBlend2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertProtrusionBlend( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal TessToleranceFactor As System.Double, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal StartTangentLength As System.Double, _  
    ByVal EndTangentLength As System.Double, _  
    ByVal StartTangentDir As System.Boolean, _  
    ByVal EndTangentDir As System.Boolean, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim TessToleranceFactor As System.Double  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim StartTangentLength As System.Double  
Dim EndTangentLength As System.Double  
Dim StartTangentDir As System.Boolean  
Dim EndTangentDir As System.Boolean  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short
```



```

Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.InsertProtrusionBlend(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,
StartMatchingType, EndMatchingType, StartTangentLength, EndTangentLength, StartTangentDir,
EndTangentDir, IsThinBody, Thickness1, Thickness2, ThinType, Merge, UseFeatScope, UseAutoSelect)

```

C#

```

Feature InsertProtrusionBlend(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.double StartTangentLength,
    System.double EndTangentLength,
    System.bool StartTangentDir,
    System.bool EndTangentDir,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ InsertProtrusionBlend(
& System.bool Closed,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.double TessToleranceFactor,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.double StartTangentLength,
& System.double EndTangentLength,
& System.bool StartTangentDir,
& System.bool EndTangentDir,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.bool Merge,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

True closes the loft, false leaves the loft open; if true and you selected less than three profiles, any selected guide curves must be closed curves

KeepTangency

True maintains the tangency as seen in the section curves, false does not

If the section curves are tangent, then you have the option to specify whether the resulting faces are also tangent; when generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface shapes if the section curves exhibit these characteristics

ForceNonRational

True obtains smoother surfaces, false does not

TessToleranceFactor

Factor that controls the number of intermediate sections used for loft with centerline; the default value is 1.0; the greater the value, the more intermediate sections are created

StartMatchingType

Tangency type at the start profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to a selected vector

- 3 = tangency to all the adjacent faces sharing an edge with the start profile
- 4 = tangent to some of the selected faces sharing an edge with the start profile (not available)

EndMatchingType

Tangency type at the end profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to a selected vector
- 3 = tangency to all the adjacent faces sharing an edge with the start profile
- 4 = tangent to some of the selected faces sharing an edge with the start profile (not available)

StartTangentLength

Start tangent length

EndTangentLength

End tangent length

StartTangentDir

True is one direction, false is the opposite

EndTangentDir

True is one direction, false is the opposite

IsThinBody

True if this feature is a thin body, false if it is not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type:

- 0 = One direction
- 1 = One direction reverse
- 2 = Mid-plane
- 3 = Two direction

Merge

True merges the results in a multibody part, false does not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertProtrusionBlend](#).

Remarks

Selection of guide curves and centerline is optional. However, you must select the profiles in an order consistent with the desired direction of the loft. Because a solid is being created, the section profiles must be closed.

It is best to use guide curves, especially when you select profiles in the FeatureManager design tree.

You can use any number of profiles; however, if you select only one profile, then any selected guide curves must be closed curves.

Use [ModelDocExtension::SelectByID2](#) to select the profiles and guide curves. Set the mark for:

- profile selections to 1
- any guide curve selection to 2
- centerline selection to 4
- start tangency vector selection to 8
- start tangency faces selection to 16 (not available)
- end tangency vector selection to 32
- end tangency faces selection to 64 (not available)

NOTE: Linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertCutBlend Method](#)
[ILoftFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertProtrusionBlend2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertProtrusionBlend2 Method (IFeatureManager)

Creates a lofted body or boss from the selected profiles, centerline, and guide curves.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertProtrusionBlend2( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal TessToleranceFactor As System.Double, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal StartTangentLength As System.Double, _  
    ByVal EndTangentLength As System.Double, _  
    ByVal StartTangentDir As System.Boolean, _  
    ByVal EndTangentDir As System.Boolean, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal GuideCurveInfluence As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim TessToleranceFactor As System.Double  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim StartTangentLength As System.Double  
Dim EndTangentLength As System.Double  
Dim StartTangentDir As System.Boolean  
Dim EndTangentDir As System.Boolean  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short  
Dim Merge As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim GuideCurveInfluence As System.Integer  
Dim value As Feature
```

```
value = instance.InsertProtrusionBlend2(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,  
StartMatchingType, EndMatchingType, StartTangentLength, EndTangentLength, StartTangentDir,  
EndTangentDir, IsThinBody, Thickness1, Thickness2, ThinType, Merge, UseFeatScope, UseAutoSelect,  
GuideCurveInfluence)
```

```

Feature InsertProtrusionBlend2(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.double StartTangentLength,
    System.double EndTangentLength,
    System.bool StartTangentDir,
    System.bool EndTangentDir,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.int GuideCurveInfluence
)

```

C++/CLI

```

Feature^ InsertProtrusionBlend2(
    & System.bool Closed,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.double TessToleranceFactor,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.double StartTangentLength,
    & System.double EndTangentLength,
    & System.bool StartTangentDir,
    & System.bool EndTangentDir,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,
    & System.short ThinType,
    & System.bool Merge,
    & System.bool UseFeatScope,
    & System.bool UseAutoSelect,
    & System.int GuideCurveInfluence
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

True closes the loft, false leaves the loft open; if true and less than three profiles are selected, then any selected guide curves must be closed curves

KeepTangency

True maintains the tangency as seen in the section curves, false does not; if the section curves are tangent, then you have the option to specify whether the resulting faces are also tangent; when generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface shapes if the section curves exhibit these characteristics

ForceNonRational

True obtains smoother surfaces, false does not

TessToleranceFactor

Factor that controls the number of intermediate sections used for loft with centerline; the default value is 1.0; the greater the value, the more intermediate sections are created

StartMatchingType

Tangency type at the start profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to a selected vector
- 3 = tangency to all the adjacent faces sharing an edge with the start profile
- 4 = tangent to some of the selected faces sharing an edge with the start profile (not available)

EndMatchingType

Tangency type at the end profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to a selected vector

- 3 = tangency to all the adjacent faces sharing an edge with the start profile
- 4 = tangent to some of the selected faces sharing an edge with the start profile (not available)

StartTangentLength

Start tangent length

EndTangentLength

End tangent length

StartTangentDir

True is one direction, false is the opposite

EndTangentDir

True is one direction, false is the opposite

IsThinBody

True if this feature is a thin body, false if it is not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type:

- 0 = One direction
- 1 = One direction reverse
- 2 = Mid-plane
- 3 = Two direction

Merge

True merges the results in a multibody part, false does not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

GuideCurveInfluence

Guide curves influence as defined in [swGuideCurveInfluence_e](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertProtrusionBlend2](#).

Example

[Insert Protrusion Blend \(VBA\)](#)
[Insert Protrusion Blend \(VB.NET\)](#)
[Insert Protrusion Blend \(C#\)](#)
[Insert Fill-surface Feature \(C#\)](#)
[Insert Fill-surface Feature \(VB.NET\)](#)
[Insert Fill-surface Feature \(VBA\)](#)
[Select Multiple Splines for Loft Guide Curves \(C#\)](#)
[Select Multiple Splines for Loft Guide Curves \(VB.NET\)](#)
[Select Multiple Splines for Loft Guide Curves \(VBA\)](#)

Remarks

Selection of guide curves and centerline is optional. However, you must select the profiles in an order consistent with the desired direction of the loft. Because a solid is being created, the section profiles must be closed.

It is best to use guide curves, especially when you select profiles in the FeatureManager design tree.

You can use any number of profiles; however, if you select only one profile, then any selected guide curves must be closed curves.

Use [IModelDocExtension::SelectByID2](#) to select the profiles and guide curves. Set the mark for:

- 1 = profile selections
- 2 = guide curve selections
- 4 = centerline selection
- 8 = start tangency vector selection
- 16 = start tangency faces selection (not available)
- 32 = end tangency vector selection
- 64 = end tangency faces selection (not available)

NOTE: Linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertCutBlend Method](#)
[ILoftFeatureData Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0